



OPEN Experimental investigation and optimization of epoxy composites reinforced with jute fiber and alumina using the Jaya ANFIS approach

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Composites reinforced with natural fibers are increasingly progressively in diverse engineering practices for their remarkable attributes, including weight reduction, high strength, cost-efficiency, biodegradability, and renewability. This research explores the mechanical attributes of epoxy composites fortified with jute fiber and alumina. Employing Response Surface Methodology (RSM) with a three-factor, three-level design, the research examines independent variables including orientation (A), size of particle (S), and weight% of particle (W) to optimize process parameters for these composites. It evaluates how these variables influence mechanical properties such as tensile strength, tensile modulus, flexural strength, flexural modulus, and water absorption capacity, aiming to enhance composite performance both economically and effectively. The combination of 0/90° fiber orientation, 106-micron particle size, and 8% weight fraction yields composites with optimized flexural and tensile properties. It is critical to effectively and affordably enhance artificial intelligence tools to establish sound mechanical properties. The Jaya algorithm, developed based on Grey-ANFIS, is utilized for process optimization, applying grey theory to establish a multi-performance index that is rigorously assessed through statistical error analysis. Based on experimental results, the study identifies optimal process variable combinations that ensure superior and enhanced multi-performance characteristics in the fabricated composites.

Keywords Composite, RSM, Mechanical properties, Natural fiber, Grey-ANFIS, Optimization, Jaya algorithm

In recent decades, natural fiber composites have garnered increasing interest from manufacturers and researchers for their wide-ranging industrial applications. These materials are favored for their low cost, lightweight properties, and eco-friendliness^{1–3}. Offering superior specific stiffness and rigidity, composites made up of bio-fibers are an exceptional alternative for specific structural and automobile applications^{4–6}. However, the inherent limitations of natural fibers bonding with matrix materials resulted in high water absorption, poor matrix bonding, low durability, and comparatively inferior mechanical and thermal properties, restricting their applications in numerous fields^{7,8}. Additionally, inadequate adherence amongst the fibers with the polymer matrix reduces mechanical performance. It is crucial to develop hybrid composites that integrate natural fibers with reinforcing particles like ceramics to overcome these challenges and enhance the desired mechanical characteristics^{9–13}. Among various natural fibers available, jute fiber stands out as a prominent and widely used fiber due to its unique attributes, including dimensional stability, abundant availability, and superior mechanical properties. Jute is considered the most significant fiber that significantly fortifies characteristics for different

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applications^{14,15}. Because of its economic viability and sustainability, it stands out as the principal alternative for extensive applications in engineering, notably in automobile and construction-led sector applications. By incorporating the jute fiber, the strengths in both compression and tension are improved^{16–20}.

Evaluating the optimal process variables is a crucial task for all manufacturers to achieve improved performance metrics. Modifying a single variable may cause the degradation of other variables chosen during the analysis of several characteristics. The issues associated with multi-criteria decision-making can be effectively mitigated through the adoption of suitable methods tailored for Multi-Criteria Decision-Making (MCDM). Grey relational analysis (GRA) stands out as a prevalent technique in the realm of multi-criteria Decision-Making^{21–22} often applied to effectively resolve complex problems involving multiple criteria. Grey relational analysis, a proven effective approach, is utilized to examine the process variables^{23–25}. Optimization of process parameters can be conducted to identify the most suitable conditions for achieving improved mechanical properties. The primary purpose of adopting an optimization technique is to enrich the projective accuracy of evolved models. Numerous models for predictions are available to generate the needed computational intelligence tools, which will be used for further predictions^{26–27}. While traditional optimization approaches are often noted for their inefficiency, they tend to produce comparatively better outputs when focused on single-aspect optimization^{14,28,29}. However, most of these methods fail when applied to multi-aspect models. Therefore, there is a need for advanced optimization methods capable of handling multiple variables simultaneously, thereby providing better-optimized variables for achieving improved performance^{30–32}. In the manufacturing of natural fiber composites, achieving desirable mechanical properties is difficult since they are dependent on a variety of input parameters. Despite the unknown and intricate correlation between the input and output variables, it is essential to figure out the appropriate values in order to get enhanced performance. Several approaches, among others, the Genetic Algorithm (GA)^{33–34} Particle Swarm Optimization (PSO)^{35–36} and Teaching Learning Based Optimization (TLBO)^{37–38} approaches, are commonly used to optimize the process variables.

The literature clearly indicates a necessity to develop and evaluate process parameters in the manufacturing of jute fiber composites utilizing both traditional and modern techniques. This involves investigating factors such as jute woven fiber (JWF), orientation (A), size of particle (S), and weight% of particle (W), while taking into account dependent variables like tensile strength, tensile modulus, weight% increase, flexural strength, and flexural modulus. Analyzing these factors demands thorough attention. Furthermore, the objective of the work is to develop a computational framework for optimization utilizing the Grey-ANFIS Jaya method.

Methodology
Response surface methodology (RSM)

Response Surface Methodology (RSM) is a unique technique employed in modeling, where data is examined through mathematical, statistical, and graphical means, taking into consideration multiple responses influenced by various input factors. Additionally, the mathematical relations can be constructed using the RSM technique. The current study incorporates the Box-Behnken Design (BBD) for experimental design approach. The design has been finalized, and the composites need to be fabricated accordingly. Table 1 shows the various levels among the process parameters used for this investigation. Utilizing a Design of Experiments (DOE) tool, the study identifies the potential sample combinations for different parameters and their corresponding levels, as detailed in Table 2.

Materials

The materials JWF, epoxy resin (LY-556), and hardener (XY-54) are acquired from Vruksha Composites located in Guntur, Andhra Pradesh, and the required alumina particles (Al₂O₃) are to be procured from Trichy-based Nice Chemicals (Sigma-Aldrich). Subsequently, the JWF is tailored to match the mold's dimensions. A mold box measuring 300 × 300 × 3 mm³ is used for fabricating composites of JWF and ceramic particles. The JWF composite samples are manufactured to meet the specified dimensions. The molding process commences with the resin and hardener being blended in the proper proportions and poured into the mold to form the JWF composite samples. Following the closure of the mold, compression is realized by applying force to the upper section of the punch plate. The flow process of the sample preparation is shown in the Fig. 1a along the compression molding machine setup shown in Fig. 1b.

As represented in Figs. 2a, b the specimens for the required flexural and tensile tests are prepared as per the ASTM-D-3039(15) & ASTM-D-790(15) and the testing was done in an Instron machine made of 8801. Similarly, ASTM-D-570 (10) standards are followed to perform the water absorption test.

Results and discussion

The composites were fabricated by investigating multiple combinations through the application of RSM, specifically employing the BBD technique. Fifteen composite samples were fabricated and then tested

Variables/Levels	Low	Medium	High
JWF orientation (A)	0–90	30–60	45–45
Size of particle (microns) (S)	53	75	106
Weight% of particle (W)	4	8	12

Table 1. Different levels of JWF orientation (A), size of particle (S) and weight% of particle(W).

Factors/ Exp. No.	A		S		W	
	Coded	Orientation	Coded	Particle size (μm)	Coded	Weight% (g)
1	-1	0/90	-1	53	0	8
2	1	45/45	-1	53	0	8
3	-1	0/90	1	106	0	8
4	1	45/45	1	106	0	8
5	-1	0/90	0	75	-1	4
6	1	45/45	0	75	-1	4
7	-1	0/90	0	75	1	12
8	1	45/45	0	75	1	12
9	0	30/60	-1	53	-1	4
10	0	30/60	1	106	-1	4
11	0	30/60	-1	53	1	12
12	0	30/60	1	106	1	12
13	0	30/60	0	75	0	8
14	0	30/60	0	75	0	8
15	0	30/60	0	75	0	8

Table 2. Experimental combinations of JWF composite samples with coded/uncoded values.

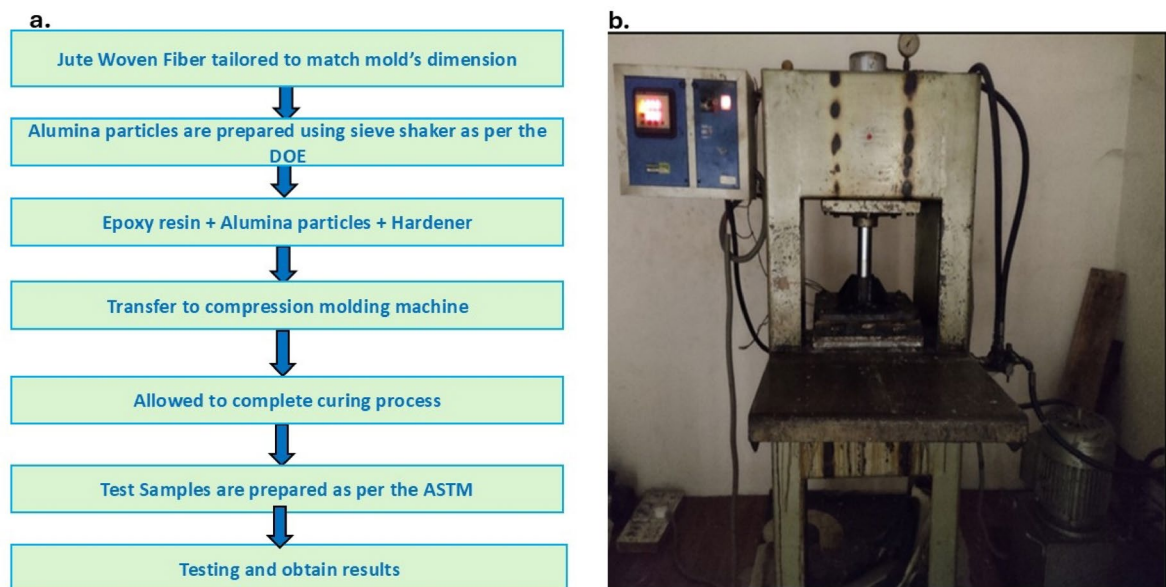


Fig. 1. (a) Process flowchart for sample preparation, (b) Compression molding machine setup.

mechanically to assess their different mechanical properties. This section discusses how the combination of processes affects the desirable mechanical characteristics.

Impact of A/S/W on tensile characteristics

Figures 3a,b demonstrate the outcome correlation among process factors on the tensile stress and tensile modulus of the fiber composites, respectively. The analysis reveals that both the properties are maximized at minimum levels of JWF orientation, weight% of particle, and size of particle at maximum levels. Notably, the particle weight% is identified as the most significant process factor impacting the tensile characteristics of the fiber composites.

Impact of A/S/W on flexural characteristics

The impacts of various variables on the flexural strength and flexural modulus of the fabricated fiber composites are examined and illustrated in Figs. 4a,b. The study demonstrates that the flexural characteristics of the composites reach their maximum when the weight% of particles and JWF orientation are at low levels and the size of particles is at the maximum level. It is noted that particle weight% is the most significant factor affecting the flexural properties of the composites tested.

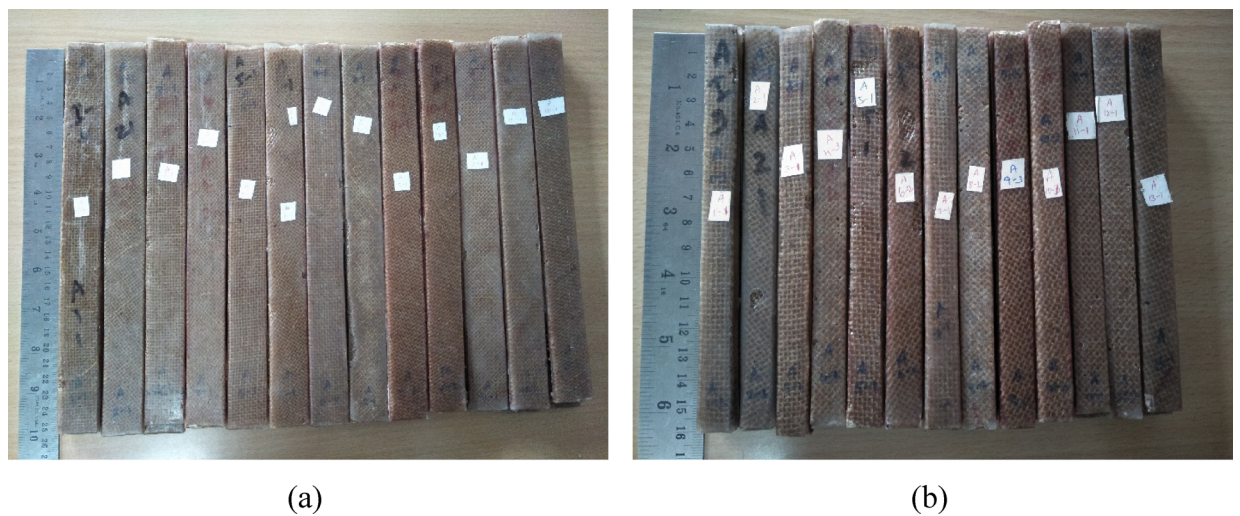


Fig. 2. Photographic images of (a) Tensile specimens, (b) Flexural specimens, and.

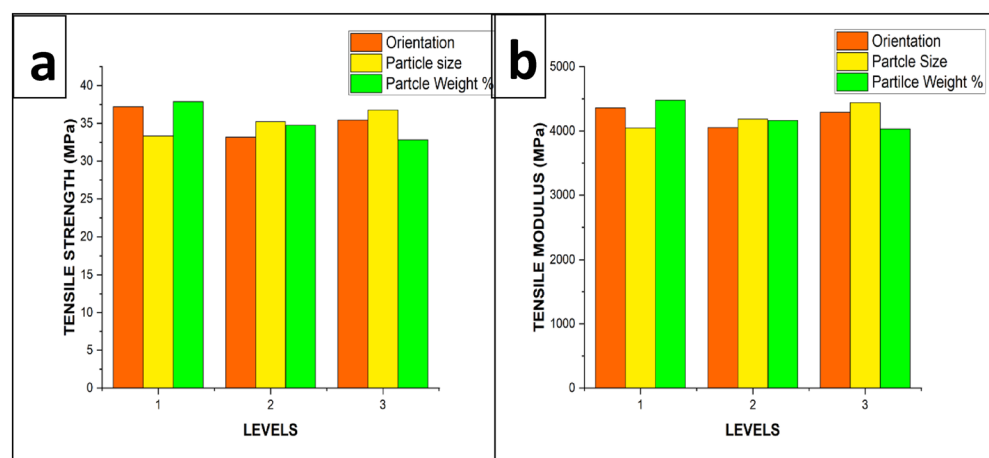


Fig. 3. The effect of process factors on (a) Tensile strength, (b) Tensile modulus.

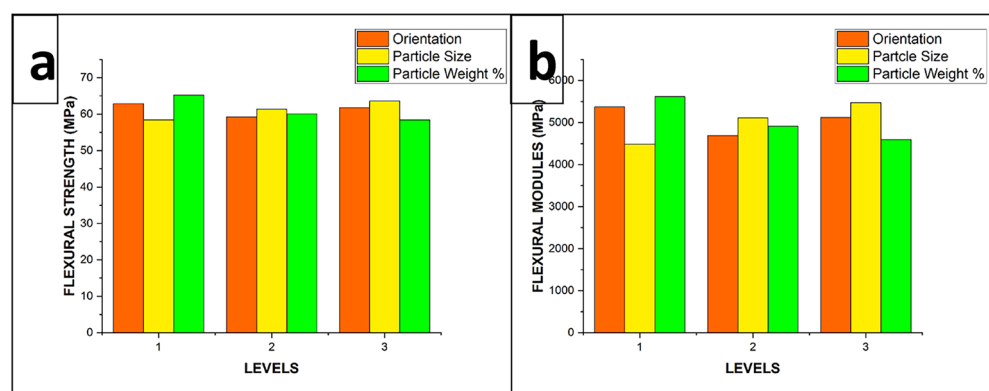


Fig. 4. Influence of process factors on (a) Flexural strength; (b) Flexural modulus.

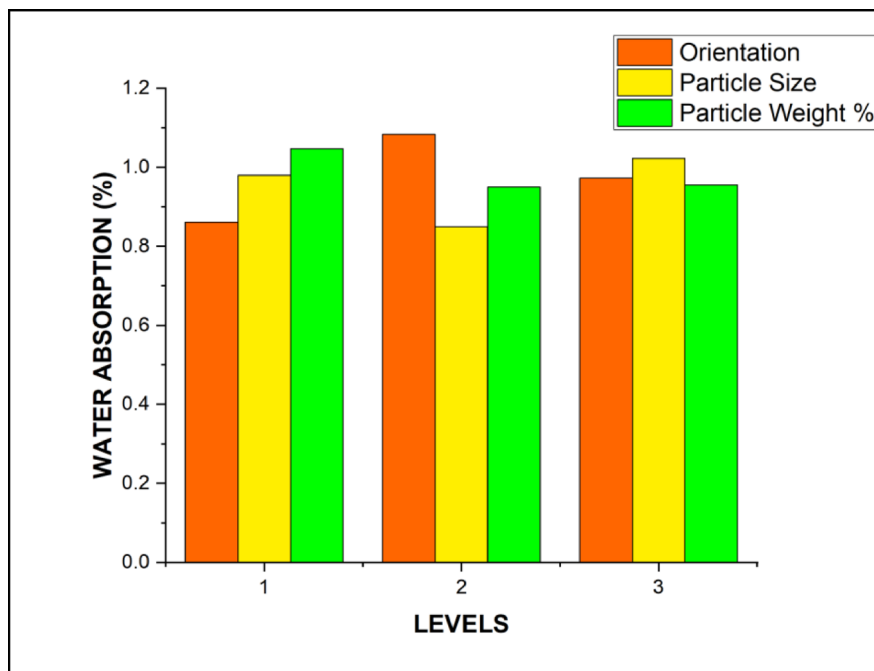


Fig. 5. Influence of factors on water absorption capacity.

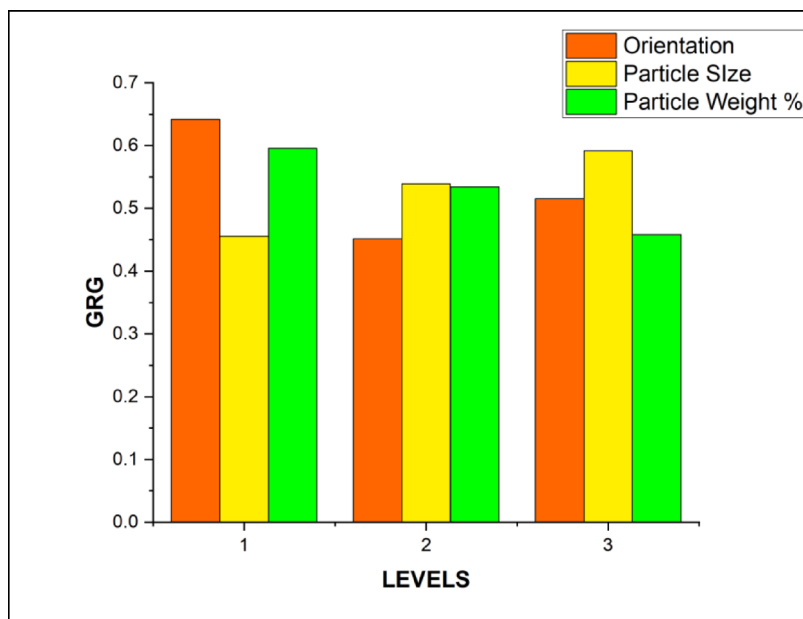


Fig. 6. Process parameters and their influence on GRG.

Impact of a/s/won the water absorption capacity

Figure 5 depicts the influence of factors on the percentage intensification in weight of produced fiber composites. The percentage increase in weight of the fiber composite is significantly higher for low levels of JWF orientation, size of particle at medium level and weight% of particle. Furthermore, particle size is identified as the key process variable determining the increase in weight of the composites produced.

Effect of factors on multi-aspects optimization

The study conducts multi-aspect optimization, and the findings are illustrated clearly in Fig. 6. It is observed that multi-objective optimization yields better outcomes at low levels of JWF orientation, particle size at the maximum level, and minimum weight% of the selected particle.

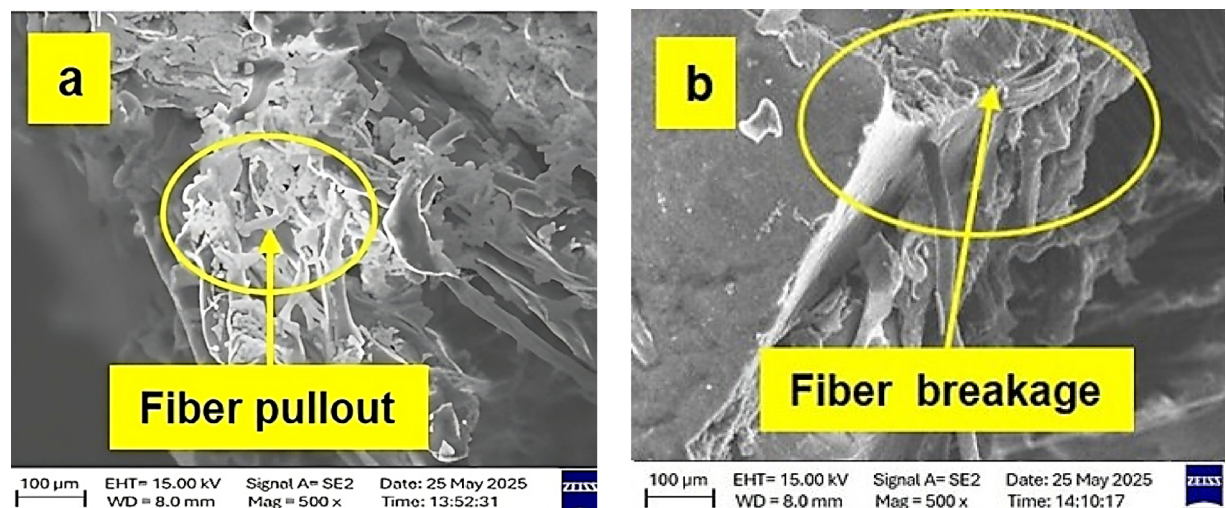


Fig. 7. Microstructural analysis of specimens after flexural testing.

Microstructural analysis

The SEM micrograph in Fig. 7a reveals the fractured surface of the jute fiber/ Al_2O_3 / epoxy composite specimen after flexural testing, showcasing a fiber pullout-dominated region. This failure mechanism indicates insufficient stress transfer from the matrix to the fiber composite reinforcement during bending. From the appearance of the pulled-out fiber, and the corresponding void it left behind, it can be inferred that the interfacial adhesion was moderate; the fiber remained intact, indicating that it did not fracture under load. The interfacial fiber matrix surface irregularity indicates the presence of some form of mechanical adhesion, albeit insufficient to counteract the bending stresses. Furthermore, microvoids or weak zones in the resin, or agglomeration of Al_2O_3 particles, may have contributed to interfacial debonding. Under flexural loading, one side of the specimen is subjected to tensile stress, and such localized defects are often associated with fiber pullout in the tensile zone.

Flexural testing of the same composite demonstrates fiber breakage, which indicates vigorous interfacial bonding as shown in Fig. 7b. At this point, the load was successfully transferred from the matrix to the fiber, and the fiber achieved its ultimate tensile limit and broke. Composites exhibiting this mode of fracture under bending are favorable, as it indicates that the matrix and composite were well wetted and adhered, with the stress highly and beneficially concentrated throughout the section. Finer matrix conformance around the fiber

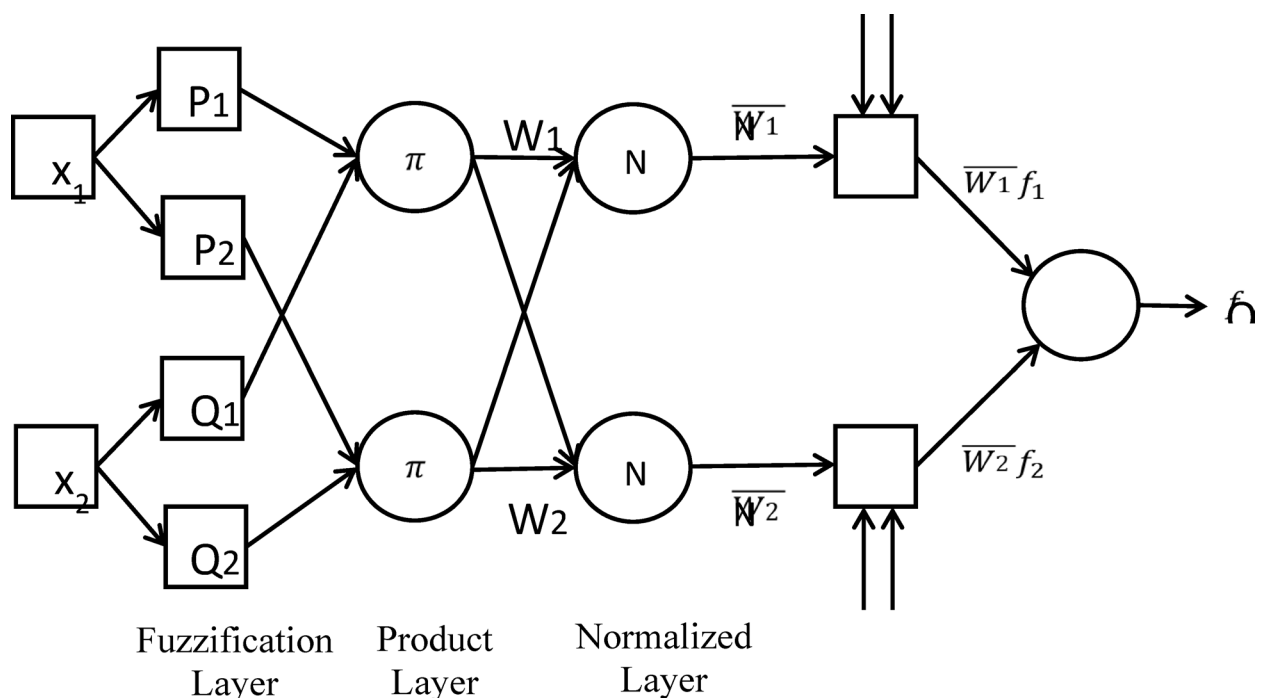


Fig. 8. Architecture of ANFIS.

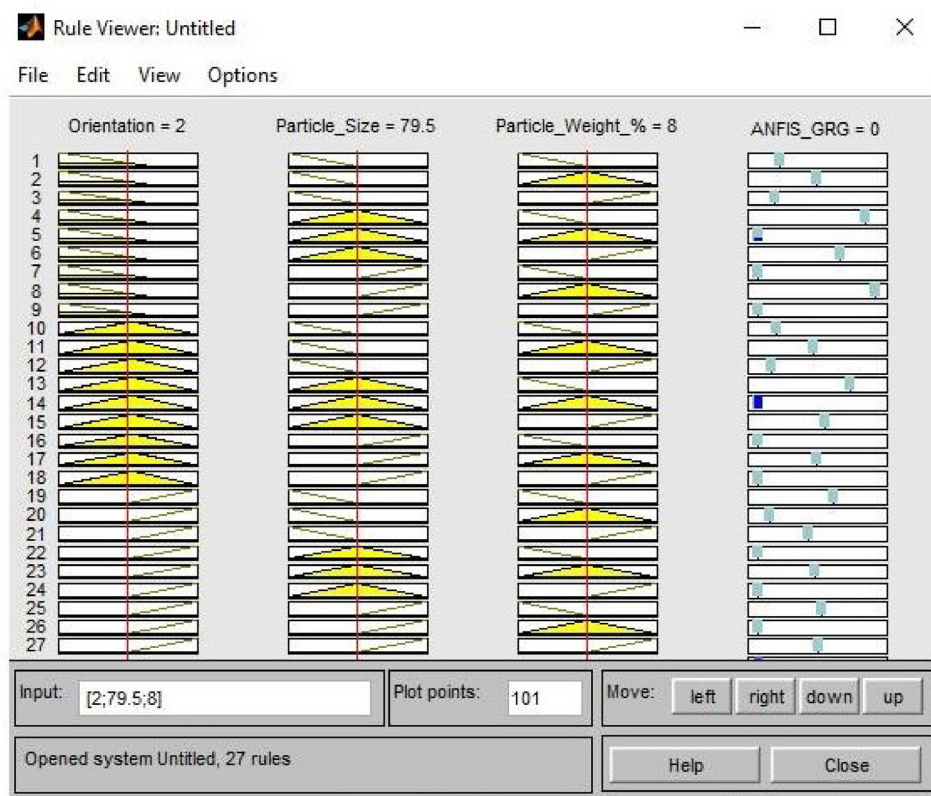


Fig. 9. Rule viewer of ANFIS.

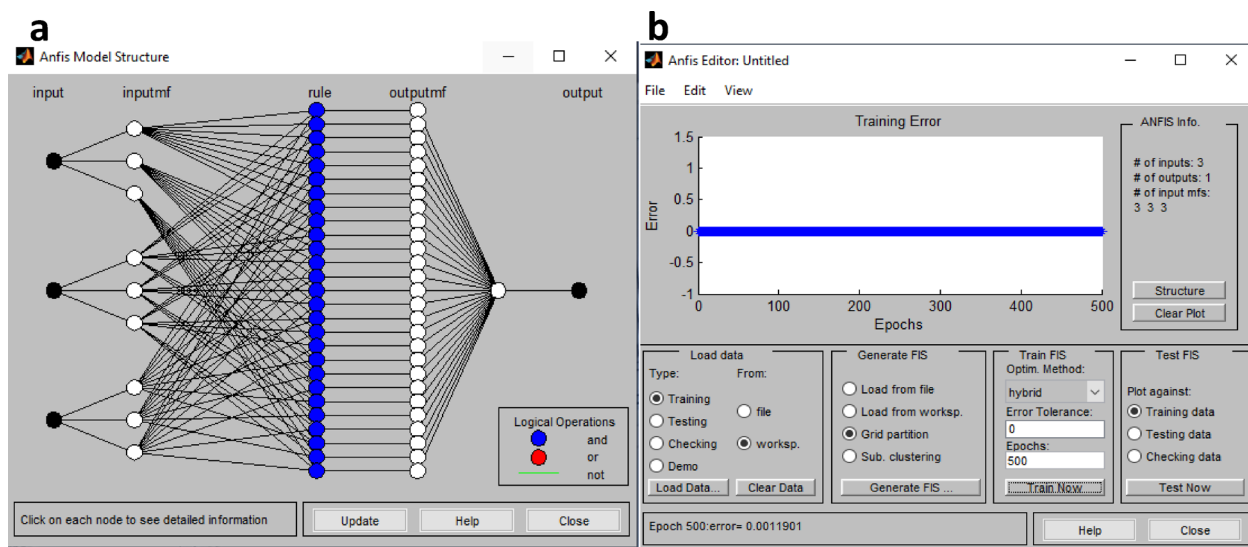


Fig. 10. (a) ANFIS structure; (b) ANFIS editor.

that had fractured indicates matrix fiber interaction enhancement, which may result from well dispersed Al_2O_3 particulates, as they increase matrix stiffness and load bearing capacity. The sample composition exhibiting a blend of fiber pullout and breakage showcases interfacial strength variability, perhaps from local fiber alignment, filler dispersion, or void content variance, all of which were utmost concerns in the Jaya-ANFIS optimization model applied in this work.

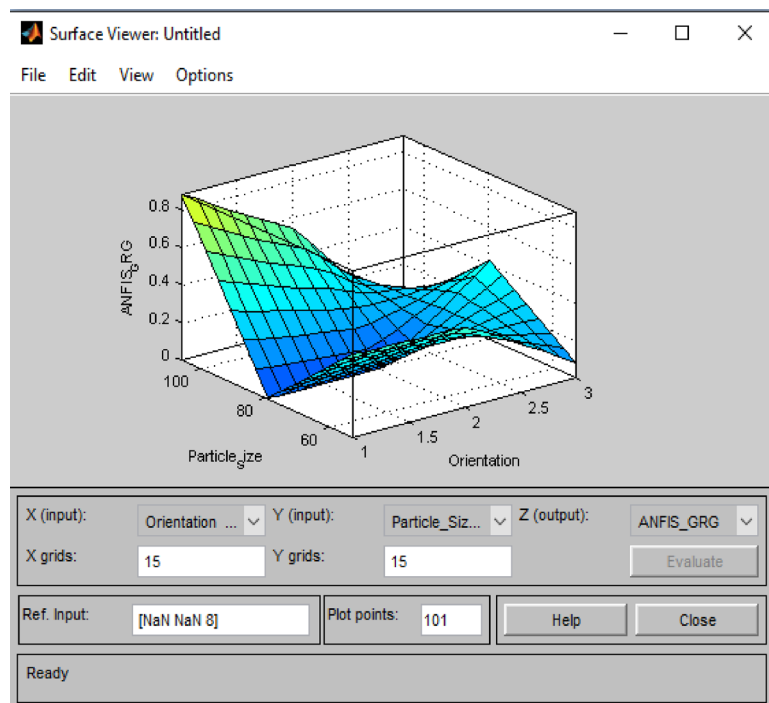


Fig. 11. Influence of A/S parameters on ANFIS-GRG.

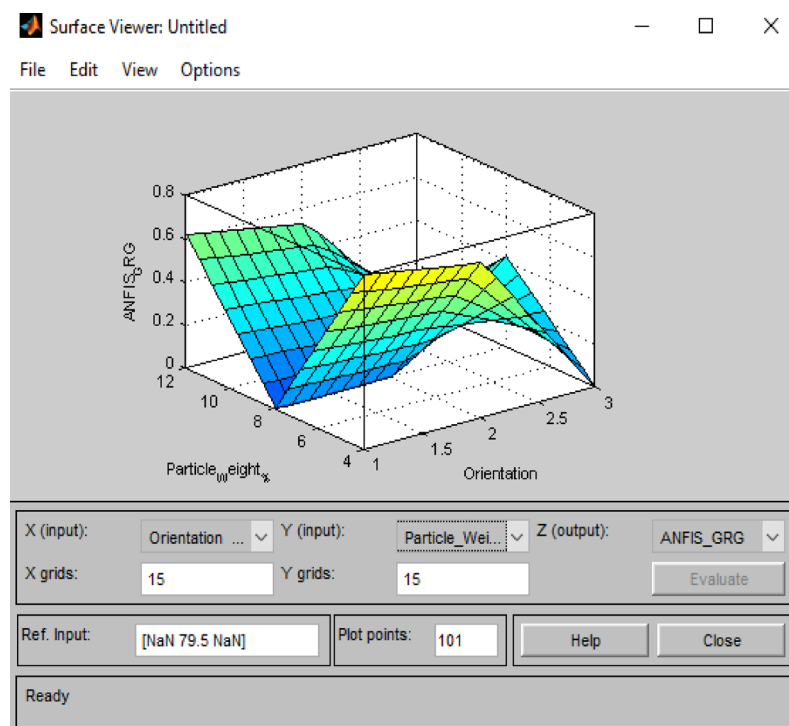


Fig. 12. Influence of A/W parameters on ANFIS-GRG.

ANFIS model and optimization

The ANFIS is a combination of the learning techniques used by artificial neural networks (ANNs) and the fuzzy inference systems (FIS) that use fuzzy logic. As a Takagi-Sugeno-type artificial intelligence framework, Fig. 8 depicts the architecture of ANFIS, which addresses complex and nonlinear problems using linear functions for the results of fuzzy norms.

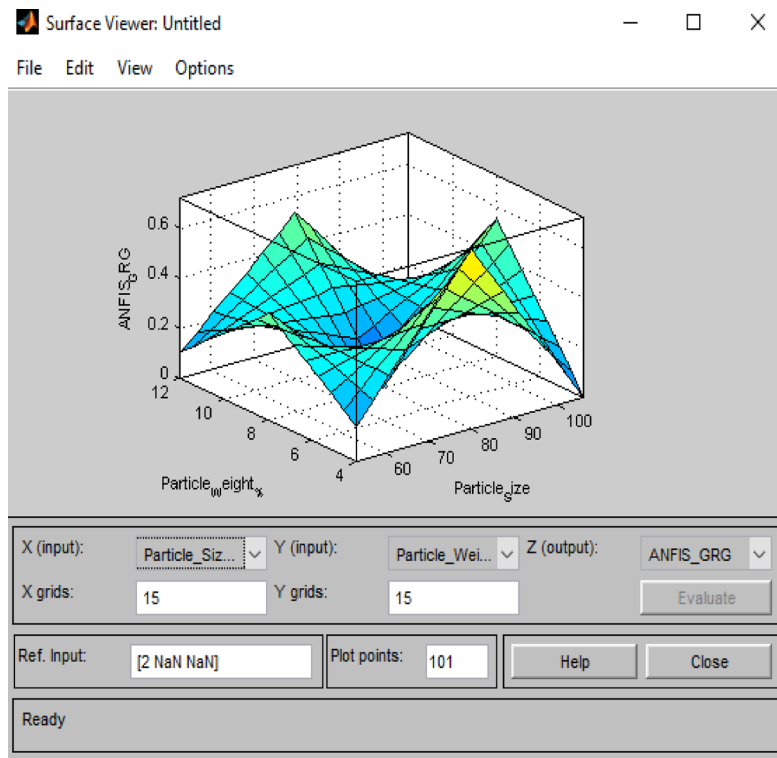


Fig. 13. Influence of S/W parameters on ANFIS-GRG.

ANFIS model	Error
MAPE	0.1324
RMSE	0.0011
Correlation coefficient	0.99

Table 3. Evaluating Grey-ANFIS model Effectiveness.

Rule 1: If x_1 is P_1 , and x_2 is Q_1 then $z_1 = a_1 \times x_1 + b_1 \times x_2 + c_1$.

Rule 2: If x_1 is P_2 , and x_2 is Q_2 then $z_2 = a_2 \times x_1 + b_2 \times x_2 + c_2$.

The variables x_1 and x_2 serve as input parameters, whereas Rule 1 and Rule 2 define z_1 and z_2 , respectively. The values of a_1, a_2, b_1, b_2, c_1 , and c_2 are determined via a learning method.

The diagram outlines ANFIS model's five-layer structural design, which includes the following layers: fuzzification, product, normalization, defuzzification, and output.

Layer-1 (Fuzzification layer)

The adaptive nodes in the fuzzification layer (*Layer 1*) use the equations below to transform the input parameters into fuzzy values.

$$O/P_i = \mu M_i(x_1); \quad i = 1, 2$$

$$O/P_j = \mu M_j(x_2); \quad j = 1, 2$$

O/P_i and O/P_j represent the outputs of nodes, while $\mu M_i(x_1)$ and $\mu M_j(x_2)$ denote the membership functions. Any proper parameterized function can be used as M 's membership function. The generalized bell function, $\mu M_i(x)$, is stated as follows:

$$\mu M_i(x_i) = \frac{1}{1 + \left| \frac{x_i - c_i}{a_i} \right|^{2b_i}}.$$

a_i, b_i, c_i are premise parameters that must be changed using a learning method.

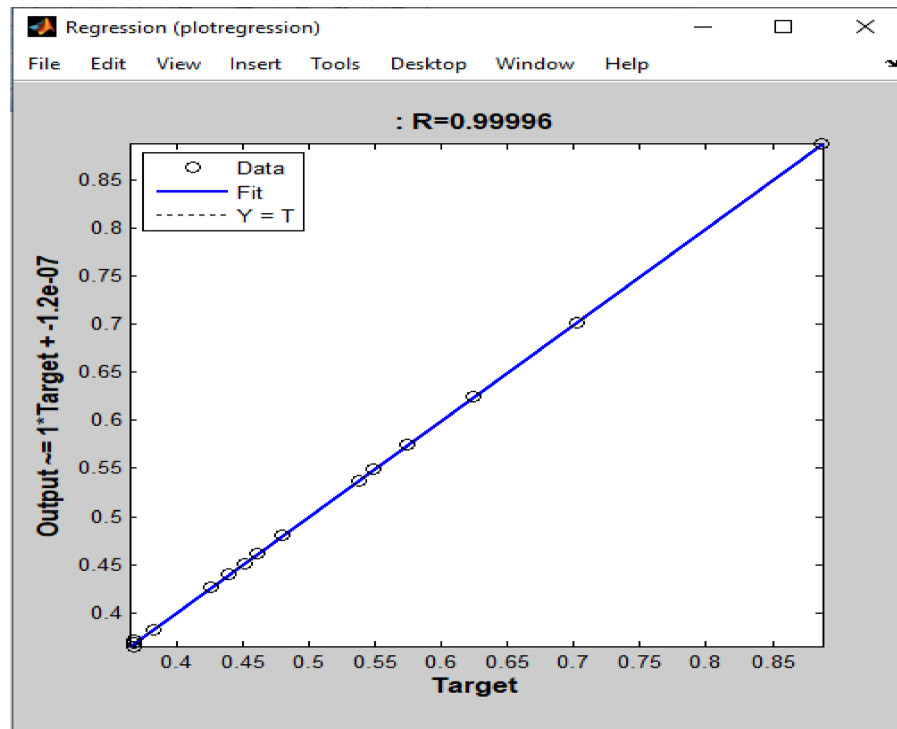


Fig. 14. Comparison between experimental GRG and ANFIS-generated GRG using regression plot.

Layer-2 (Product layer)

The function of the node in this layer is to figure out the rule's firing strength using the following formula.

$$\omega_i = \mu_{M_i}(x_1) \mu_{M_i}(x_2), \quad i = 1, 2$$

Here ω_i represents the firing strength of the i^{th} rule.

Normalized layer (Layer 3)

The firing intensity of the system's i^{th} node is normalized using the equation below.

$$\bar{\omega}_i = \frac{\omega_i}{\omega_1 + \omega_2}$$

,

Layer-3 (Defuzzification layer)

In defuzzification, fuzzy values are converted into precise parameter values. This layer consists of adaptive nodes, each with an associated function.

$$\bar{\omega}_i f_i = \bar{\omega}_i (p_i x + q_i y + r_i)$$

The following parameters p_i , q_i , and r_i are also changed using a learning method.

Layer-5 (Output layer)

In this layer, a node determines its output data by performing the sum of all input signals as outlined below:

$$Output = \sum_i \bar{\omega}_i f_i = \frac{\sum_i \omega_i f_i}{\sum_i \omega_i}$$

The ANFIS model built in this work has three input variables and one output variable, which have been trained to predict certain outcomes. The 'trimf' membership functions, applied in the model, lead to the automatic generation of 27 rules that depend on the given data. The rules generated can be seen in the rule viewer, which is depicted in Fig. 9. Grey concept, a classical MCDM method, is utilized to compute Grey Relational Coefficients (GRC) for use in the finalized model. The experimental results are then normalized to further enhance the ANFIS structure. Visual representations of the ANFIS model structure and editor are presented in Figs. 10a, b respectively. By applying the ANFIS-GRG model, the multi-performance index can be forecasted effectively.

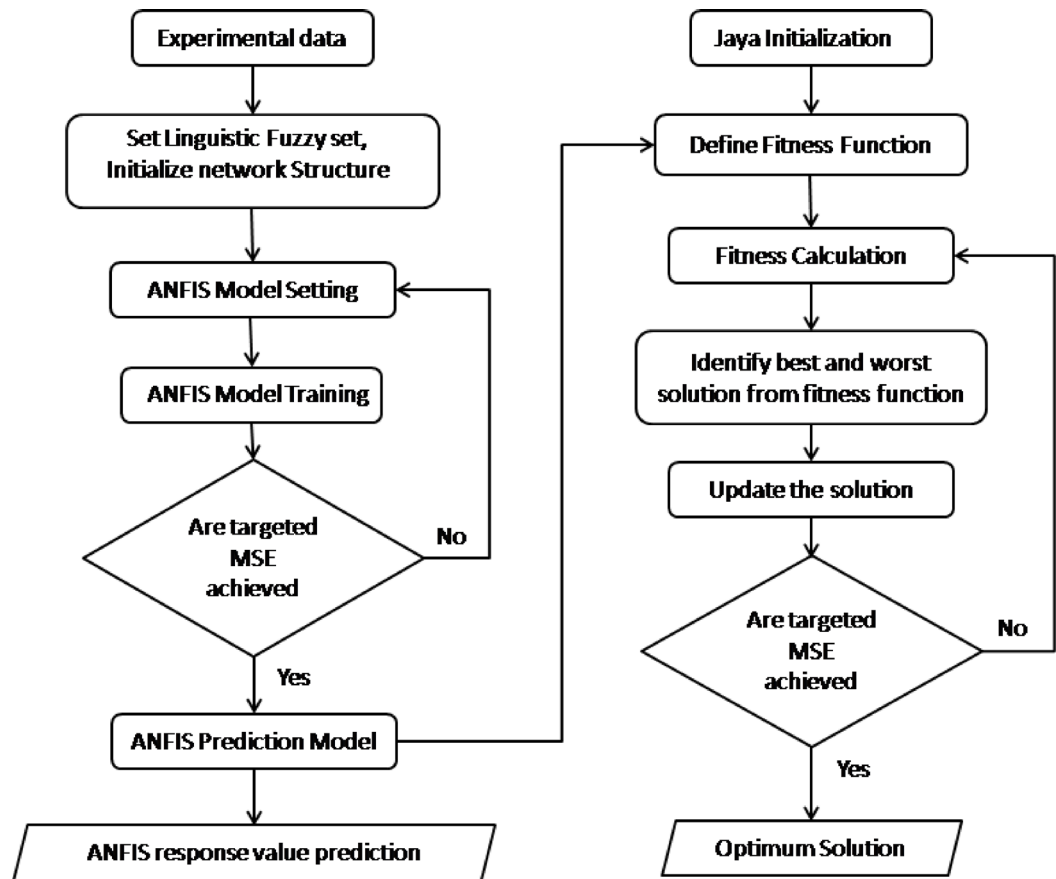


Fig. 15. Schematic flow of the Grey-ANFIS based Jaya optimization process.

Evaluation of various selected parameters on ANFIS-GRG

The surface plots illustrating the Grey Relational Grade (GRG) as a function of orientation and particle size are presented in Fig. 11. The plots indicate that an improved GRG is attained with a combination of low orientation levels and high particle size levels. Additionally, the surface plots for GRG as a function of orientation and particle weight% are depicted in Fig. 12. These plots demonstrate that lower levels of both orientation and particle weight% result in a more significantly enhanced GRG. The surface plots depicting the grey relational grade (GRG) in relation to the size of the particle and weight% of the particle are unveiled in Fig. 13. The graphical representation indicates that a higher level of particle size combined with a lower level of particle weight% yields an exceptional and refined ANFIS-GRG.

Evaluation of the performance of the ANFIS model

The error quantifies the deviation between experimental results and the model's predicted outcomes. Using Eqs. (1–3), several measures such as the Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and correlation coefficient are produced to assess the effectiveness of the established ANFIS model.

$$\text{MAPE}(\%) = \frac{1}{n} \sum_{i=1}^n \frac{A_v - F_v}{A_v} * 100 \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (A_v - F_v)^2} \quad (2)$$

$$\text{Correlation coefficient} - R^2 = 1 - \frac{\sum_{m=1}^n (F_v - A_v)^2}{\sum_{m=1}^n (A_v)^2} \quad (3)$$

Where,

' A_v ' - actual values, ' F_v ' - forecasted value and ' n ' is the samples observed in a study.

The deviation between the ANFIS model's predicted outcomes for JWF composites and experimental results is analyzed, showing a close correlation. The performance of the model is evaluated and summarized in Table 3 using Eqs. (1–3), and Fig. 14 depicts the regression curve.

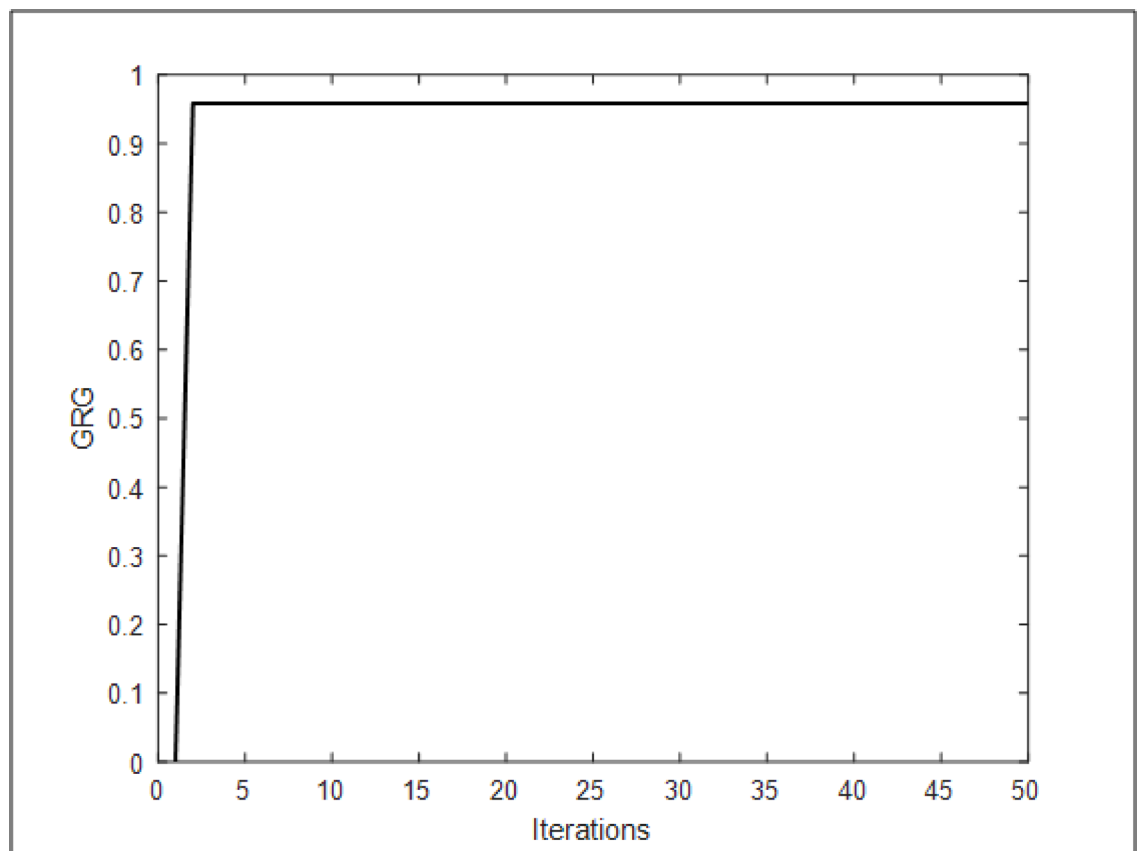


Fig. 16. JAYA algorithm – convergence graph.

Parameter	Grey-RSM	Jaya-ANFIS
JWF orientation	0/90	0/90
Size of particles (microns)	106	106
Weight% of particle	8	4
Optimum GRG	0.8877	0.9586
Improvement %	7.99	

Table 4. Comparing optimal process variables achieved through Grey-RSM and Jaya-ANFIS methods.

JAYA-ANFIS optimization model

The Jaya Algorithm (JA), introduced by Ravipudi Venkata Rao³⁹ has garnered significant interest from diverse research areas owing to its remarkable features, which are conceptually simple and user-friendly. During the initial search, the algorithm is not relying on derivative insights; additionally, the algorithm is a parameter-free method that is adaptive, versatile, sound, and comprehensive³¹. In the current work, the Jaya method was combined with the Grey-ANFIS model, which is based on grey theory and is used to identify the most ideally selected variables for increasing the multi-performance index. The improved accuracy of the ANFIS modelling assists the JA to search for optimal solutions more efficiently, which could lead to more precise results. Figure 15 illustrates the method for constructing the Grey-ANFIS-based JA.

The developed algorithm identifies the optimal combination of selected parameters to accomplish exceptional multi-performance in JWF composites with a 0/90 orientation, a particle size of 106 microns, and 4% particle weight, achieving a finest fitness metric of 0.9586. Figure 16 depicts the evolution of the Jaya algorithm for optimizing multi-performance in tested JWF composites.

The comparison in Table 4 illustrates the optimal process variables obtained through both the Grey-RSM and Jaya-ANFIS algorithms. The findings highlight that the Jaya-ANFIS algorithm ensures the highest precision in establishing the optimal parameters.

Conclusions

In this research, a three-tier and three-factor model is employed using the RSM-BBD method. The composites are produced using a compression molding machine, and their mechanical properties, such as tensile and

flexural characteristics, in addition to their water absorption capacity, are also assessed. The study highlights that fiber orientation is pivotal as a procedural factor, with particle weight% also significantly impacting mechanical behavior. Furthermore, a predictive model is developed using a hybrid Grey-ANFIS technique, demonstrating enhanced accuracy in forecasting performance metrics. The study identifies optimal process parameters for maximizing multi-performance in manufacturing jute fiber composites using the Jaya-ANFIS algorithm, achieving a peak fitness value of 0.9586 with a 0/90 fiber orientation, 106 micron particle size, and 4% particle weight. This optimization resulted in significantly enhanced performance of the composites, achieving a tensile strength of 45.786 MPa, a flexural strength of 74.507 MPa, and a water absorption capacity of 1.19%. The application of the Jaya-ANFIS algorithm facilitates the determination of the most effective combination of process variables for these composites, offering superior outcomes compared to conventional Grey-RSM approaches.

Data availability

The necessary data used in the study are presented in the main manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

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The authors consent to the publication of this manuscript.

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The authors affirm that this manuscript provides a truthful, accurate, transparent research account. No significant aspects of the study have been omitted, and any deviations from the original study plan (and, if applicable, registration) have been fully explained.

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